

Closed loop synchronous buck converter 60W design

1. **Objective:** Simulation and analysis of a Closed-loop 60W synchronous buck converter in LTspice incorporating practical parasitic elements.
2. **Specifications:**
 - 2.1 Input voltage (V_{in}): 42 V
 - 2.2 Output voltage (V_{out}): 12 V
 - 2.3 Output Current (I_{out}): 5 A
 - 2.4 Output power (P_{out}): 60 W
 - 2.5 Switching frequency (F_{sw}): 200 kHz
 - 2.6 MOSFET: BSC190N12NS3(Infineon)
 - 2.7 Load resistance (R_1): 2.4Ω
 - 2.8 Input source resistance: $100m\Omega$
 - 2.9 Input parasitic inductance: $200nH$
 - 2.10 Inductor DCR: $10m\Omega$
 - 2.11 Capacitor ESR: $30m\Omega$
 - 2.12 Output capacitor: $22\mu F$
 - 2.13 VCC(V_5): 12V
 - 2.14 VREF(V_3): PULSE (0 3.3 0 2m 10n 20m 20m)V
 - 2.15 VCC & VREF internal resistance: $10m\Omega$
 - 2.16 Feedback resistance (Top side): $26.36K\Omega$
 - 2.17 Feedback resistance (bottom side): $10K\Omega$
 - 2.18 High side dead time control(B_1): $V = \text{delay}(\text{if}(V(PWM_OUT) > 6, 9.5, 0), 5n)$
 - 2.19 Low side dead time control(B_2): $V = \text{delay}(\text{if}(V(PWM_OUT) < 6, 9.5, 0), 30n)$
 - 2.20 Error amplifier type: Active type – II compensation network ($R_5=27K\Omega$, $R_4=40\Omega$, $C_3=1.2\mu F$, $C_4=62pF$).

3. Design calculation:

3.1 Baseline power stage reference: The passive power stage component values and switching frequency were retained from previously developed asynchronous buck converter design, as the required electrical specifications remained unchanged. While the inductor design rationale, CCM boundary verification and passive component selection are documented in the companion synchronous buck converter report.

3.2 Ideal Duty Cycle and Initial High-Side ON-Time Estimation: For an ideal buck converter operating in continuous conduction mode, the duty cycle is given by:

$$D_{ideal} = \frac{V_{out}}{V_{in}}$$

Substituting the design values: $D_{\text{ideal}} = \frac{12}{42} = 0.2857$

The switching period corresponding to 200kHz operation is: $T_s = \frac{1}{f_s} = \frac{1}{200000} = 5\mu\text{s}$

Therefore, the Ideal theoretical high-side MOSFET ON-Time is:

$$T_{\text{ON,ideal}} = D_{\text{ideal}} * T_s = 0.2857 * 5 = 1.428 \mu\text{s}$$

In a practical simulation, this ideal 1.428 μs ON-time serves strictly as a starting baseline. Because our system incorporates real world parasitic elements such as Inductor DCR and capacitor ESR and internal MOSFET conduction channel resistances ($R_{\text{DS(on)}}$) – inherent voltage drops will occur across the power loop under a full 5A load. To overcome this localized electrical losses and prevent output voltage sagging below the 12V target, the closed-loop type – II error amplifier will automatically scale the actual operating duty cycle slightly upward. This dynamic correction ensures stable steady-state regulation at the cost of minor thermal power losses analysed later in this report.

3.3 Dynamic Duty Cycle and ON-Time Compensation Under Non-Ideal Conditions:

In an ideal buck converter operating in Continuous Conduction Mode (CCM), the output voltage depends strictly on the ideal duty ratio ($D = \frac{V_{\text{out}}}{V_{\text{in}}}$). However, a physical circuit contains non-ideal parasitic elements that introduce internal voltage drops under a full 5A load. These loss channels include:

- MOSFET Static Drain-Source On-Resistance ($R_{\text{DS(on)}}$)
- Inductor Direct Current Resistance ($\text{DCR} = 10\text{m}\Omega$)
- Capacitor Equivalent Series Resistance ($\text{ESR} = 30\text{m}\Omega$)
- High-side and low-side switching transition and dead-time body diode losses

If this converter were operated open-loop at the calculated ideal high-side ON-time of 1.428 μs , these cumulative energy losses would cause the output voltage to drop significantly below the 12V target.

Because this design implements a closed-loop system, the active Type-II compensation network continuously monitors the output voltage through the feedback divider network. When the parasitic elements cause the voltage to drop, the error amplifier automatically scales its control voltage output. This corrective action automatically increases the operational pulse-width modulation (PWM) duty cycle from the ideal theoretical value (28.57%) to a practical operating value of approximately 30.12%.

As a result of this automatic loop regulation, the high-side MOSFET ON-time dynamically settles at 1.506 μs during steady-state simulation (verified via cursor measurements at 9.994ms. This automated tracking successfully stabilizes the converter output at the targeted regulation line, completely overcoming the physical losses of the system without manual user adjustment.

3.4 Dead-Time Selection and Behavioural Syntax Derivation: To prevent simultaneous conduction of the high-side and low-side MOSFET a non-overlap dead-time safety window is mathematically enforced using LTspice arbitrary behavioural voltage sources (B-sources). The setup implements a fixed switching period ($T_s = 5\mu s$, $f_{sw} = 200KHz$) referenced to a central PWM control node(V_{PWM_OUT}).

Behavioural implementation equations are:

B1(High side): $V = \text{delay}(\text{if}(V(PWM_OUT) > 6.9.5, 0), 5n)$

B2(Low side): $V = \text{delay}(\text{if}(V(PWM_OUT) < 6.9.5, 0), 30n)$

This behavioural engine determines the physical switching delays relative to the transition edges of the central V_{PWM_OUT} control loop voltage:

Selected Dead-Time: $T_{dead} = 25ns$

- High side Turn-on delay ($T_{delay,HS}$): Triggered when the command signal climbs past the 6V threshold turning the high-side switch on after the blanking period:

$$T_{delay,HS} = 5ns$$

- Low-side Turn-on delay($T_{delay,LS}$): Triggered when the control voltage drops below the 6V threshold, indicating high-side turn-off completion:

$$T_{delay,LS} = 30ns$$

Through this conditional behavioural framework, a symmetrical 25ns delay vector is executed at both rising and falling edges of switching phase. This prevents current path overlapping across the power rails while optimizing the synchronous rectification conduction interval.

This ensured symmetrical dead-time at both switching transitions.

3.5 PWM Generation and Closed-Loop Control Architecture: To regulate the converter output precisely at 12V under dynamic line and load conditions, a closed-loop negative feedback structure is modelled. This section details the mathematical generation of the central pulse-width modulation signal (V_{PWM_OUT}) that acts as the timing trigger for the behavioural gate drivers.

3.5.1 Voltage sensing, Bias power and error generation: The controller logic requires a stable auxiliary rail and an absolute reference voltage to process feedback signals accurately:

- Bias supply voltage(Vcc): Set to 12V, acting as the power supply for the internal error amplifier and providing the 9.5V amplitude output voltage limits for the behavioral gate signals.
- Reference voltage(Vref): Set to a pulsating value of 3.3V having Vinitial = 0V, Von = 3.3V, Tdelay = 0s, Trise = 2ms, Tfall = 10ns Ton & Tperiod = 20ms establishing the high-precision target for the feedback loop. The reason behind this pulsating value is to providing a soft start to a system.

The actual output voltage (Vout) is scaled down via a behavioral resistor divider network ration ($H = 10K / (10K + 26.36K) = 0.275$) to match the reference target ($12 * 0.275 = 3.3V$). The error generation block continuously calculates the instantaneous tracking deviation:

$$V_{\text{feedback}}(t) = 0.275 * V_{\text{out}}(t)$$

$$V_{\text{error}}(t) = 3.3V - V_{\text{feedback}}(t)$$

3.5.2 Feedback voltage divider parameter: To scale the nominal 12V output down to match the internal 3.3V Control loop reference(Vref), an analytical voltage divider derivation is executed. Setting the low-side grounding resistor(R_{low}) to a standard engineering baseline of $10K\Omega$ to balance noise immunity against thermal leakage current, the high-side feedback resistor(R_{high}) is calculated using the standard non-inverting scaling:

$$V_{\text{ref}} = V_{\text{out}} * \left(\frac{R_{\text{low}}}{R_{\text{high}} + R_{\text{low}}} \right)$$

Rearranging the formula to solve target high-side resistor (R_{high}):

$$R_{\text{high}} = R_{\text{low}} * \left(\frac{V_{\text{out}}}{V_{\text{ref}}} - 1 \right)$$

$$R_{\text{high}} = 10K * \left(\frac{12}{3.3} - 1 \right) = 26.363K\Omega$$

This precise mathematical derivation yields a feedback scaling ratio of $H = 0.275$, ensuring the error generation block continuously calculates tracking derivations with zero steady-state offset error:

3.5.3 The Behavioral PWM Comparator Logic: The central switching command(V_{PWM_OUT}), is generated by comparing a dynamic control error voltage (V_{CTRL}) against an independent, high-frequency sawtooth ramp waveform (V_{RAMP}) oscillating at the converter's switching frequency (200KHz).

The mathematical condition modelled inside the behavioral PWM block is defined as follows:

$$\text{If } V_{CTRL}(t) > V_{RAMP}(t) = V_{PWM_OUT} = 9.5V$$

$$\text{If } V_{CTRL}(t) \leq V_{RAMP}(t) = V_{PWM_OUT} = 0V$$

This translates into a high-frequency square wave the PWM_OUT node, whose pulse width shifts proportionally with the magnitude of V_{CTRL} .

3.5.4 Closed-loop feedback compensator structure: The control voltage(V_{CTRL}) is continuously updated by a proportional-integral(PI) error amplifier network modelled via behavioural voltage expressions. The sensing loop functions through three distinct stages.

- **Voltage sensing:** The actual output voltage (V_{out}) is scaled via a behavioural feedback ratio(H) to yield $V_{feedback}(t) = H * V_{out}(t)$.
- **Error amplification:** The scaled feedback signal compared against a stable internal DC reference voltage(V_{ref}) to compute the transient error voltage:

$$V_{error}(t) = V_{ref} - V_{feedback}(t)$$

- **PI control action:** The mathematical engine applies proportional and integral gains to generate the instantaneous control signal feeding the PWM stage:

$$V_{CTRL}(t) = K_p * V_{error}(t) + K_i \int_0^t V_{error}(t) dt$$

3.5.5 Type-2 Compensation design: A type 2 compensator is designed to stabilize the closed-loop synchronous buck converter and achieve accurate output voltage regulation.

Step 1: Calculate LC Resonant Frequency

$$L = 33 \mu H, C = 22 \mu F$$

The LC resonant frequency is:

$$F_{LC} = 1 / [2\pi\sqrt{LC}]$$

Substituting the values:

$$F_{LC} = 1 / [2\pi\sqrt{(33 \times 10^{-6} \times 22 \times 10^{-6})}] = 5.9 \text{ kHz}$$

Step 2: Select Compensator Zero Frequency

The compensator zero is initially chosen at approximately one-third of the LC resonant frequency.

$$F_z = F_{LC} / 3$$

$$F_z = 5900 / 3 = 1966.66 \approx 2\text{KHz}$$

Using:

$$R_4 = 40 \Omega$$

The compensator capacitor C3 is calculated as:

$$C_3 = 1 / (2\pi \times R_4 \times f_z)$$

$$C_3 = 1 / (2\pi \times 40 \times 2000) = 1.989\mu\text{F} \approx 1.2\mu\text{F} \text{ (Optimized value)}$$

Step 3: Select High-Frequency Pole

The compensator pole is selected at approximately half the switching frequency.

Switching frequency:

$$F_{sw} = 200 \text{ kHz}$$

$$f_p = f_{sw} / 2 = 100 \text{ kHz}$$

Using:

$$R_5 = 27 \text{ k}\Omega$$

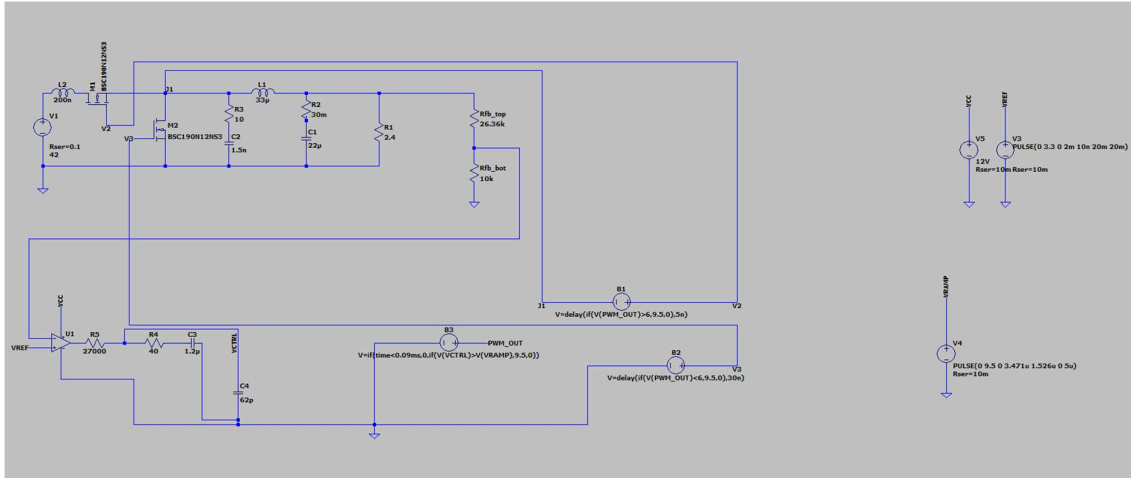
The capacitor C4 is calculated as:

$$C_4 = 1 / (2\pi \times R_5 \times f_p)$$

$$C_4 = 1 / (2\pi \times 27000 \times 100000)$$

$$C_4 = 58.95 \text{ pF} \approx 62 \text{ pF}$$

4. Circuit Diagram:



4.1 Circuit overview: The closed-loop synchronous buck converter architecture consists of two main loops: The power stage and feedback control loop. The power stage utilizes a high-side MOSFET(Q1) and a low-side synchronous MOSFET(Q2) to efficiently step down the input voltage. The output LC filter (L1 & C_{out}) smoothes the switching waveform into a stable DC voltage across the 2.4Ω load. Simultaneously, the control stage continuously samples the output voltage, compares it to a reference voltage and adjusts the switching duty cycle via PWM to correct any voltage deviations.

4.2 Circuit working and operational theory: The fundamental objective of the Closed-Loop Synchronous Buck converter is to step down a higher DC input voltage(V_{in}) to a lower, highly regulated DC output voltage(V_{out}) across the 2.4Ω load. This is achieved by utilizing high-frequency electronic switching to modulate the energy stored within Inductor(L₁) and Output capacitor(C_{out}).

4.2.1 Interval-1 High-side switch phase(Q1 ON, Q2 OFF): When the controller drives the PWM signal to a logic HIGH state, the high-side MOSFET(Q₁) transitions into full condition, acting as a closed switch. Simultaneously, the low-side MOSFET(Q₂) is driven completely OFF.

- **Current Path:** Current flows directly from the input power supply through the source-drain channel of Q₁, passes through power inductor L₁ and delivers energy simultaneously to the output filtering capacitor C_{out} and 2.4Ω load resistor.
- **Inductor Dynamics:** During this interval, the voltage across the Inductor is equivalent to the difference between the input and output voltage ($V_L = V_{in} - V_{out}$). Because this voltage is positive, the Inductor current(I_L) increases linearly. Electrical energy is converted and stored within the inductor's magnetic field, preventing sudden spikes in current to the load.

4.2.2 Interval-2 Low-side Synchronous phase (Q1 OFF, Q2 ON): When the PWM signal transitions to a logic LOW state, the controller immediately commands Q1 to turn OFF, Following a strictly timed dead-time delay-programmed to ensure both transistors are never on at the same time the low side MOSFET (Q2) is driven ON.

- **Current Path:** Because an Inductor naturally opposes any instantaneous change in current, its internal Magnetic field collapses, reversing its voltage polarity($V_L = -V_{out}$). The Inductor now acts a temporary current source. It pulls current from the system group up through the channel of Q2, through L1 and continues downstream into a 2.4Ω load.
- **Synchronous Rectification advantage:** Because a fully turned-on MOSFET exhibits an incredibly low channel resistance ($R_{DS(on)}$), the conduction voltage drop across Q2 is dramatically smaller than the typical 0.5 to 0.7V forward drop of a standard Schottky diode. At a tight 2.4Ω load where output current demands are elevated, this active rectification minimizes power dissipation(I^2R losses) and keeps thermal limits stable.

4.3 Closed-Loop feedback and Regulation Mechanism: While the power stage handles the physical conversion of energy, the closed-loop control system serves as the intelligence of the converter. Under a constant 2.4Ω load, any fluctuation in the input voltage or any parasitic losses across the components will cause the output voltage to drift from its target value. To maintain a regulated output, continuous negative feedback loop is implemented. The entire process behaves as an automated, high-speed adjustment cycle divided into four sequential operational steps:

4.3.1. Output Voltage Sensing: The circuit cannot correct a deviation without accurate tracking. A resistive voltage divider network taps into the output rail directly across the 2.4Ω load. This network acts as a scaling mechanism, taking the raw output voltage and reducing it to a lower, safe proportional signal known as the feedback voltage. This feedback voltage serves as a real-time "mirror image" of what is happening at the load.

4.3.2. Error Evaluation and Comparison: The scaled feedback voltage is routed directly into an internal error amplifier. The opposing input of this amplifier is tied to a highly stable, fixed internal reference voltage, which represents the exact target output voltage of the design. The error amplifier acts like a balance scale, constantly comparing the feedback voltage against the target reference. If the output voltage drops even a fraction of a volt due to the high current demand of the 2.4Ω load, the amplifier instantly detects the imbalance and outputs an analog error signal indicating that the output is too low.

4.3.3. Control Loop Compensation: Before the error signal is allowed to alter the switching of the MOSFETs, it passes through a dedicated compensation network consisting of filtered resistors and capacitors. A raw error signal is often too erratic forcing the circuit to react to it directly would cause the system to overcorrect, panic, and oscillate wildly. The compensation network acts as an electronic shock absorber. It smooths out the error signal, granting the converter a fast reaction time while maintaining system stability and preventing harmful voltage ringing.

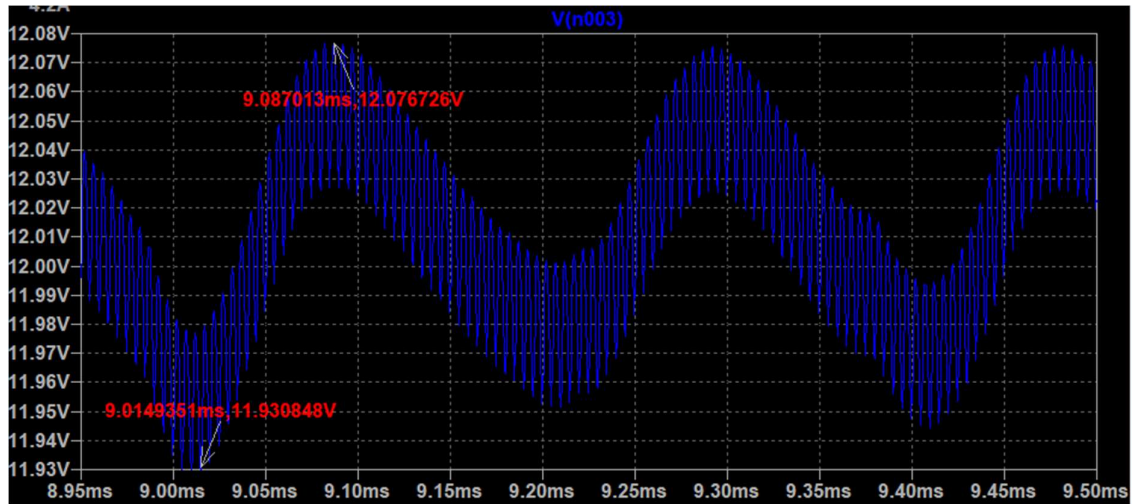
4.3.4. Pulse Width Modulation (PWM) Adjustment: The stabilized error signal is finally sent to the PWM generator, which dictates the gate control signals for the switching transistors.

- **Under-Voltage Correction:** If the output voltage sags, the error signal commands the PWM generator to widen the control pulses. This forces the high-side MOSFET to stay turned ON longer during each switching cycle, pumping more energy into the inductor and driving the output voltage back up to its target.
- **Over-Voltage Correction:** Conversely, if the output voltage rises too high, the pulses are narrowed. The high-side MOSFET shuts off sooner, throttling the energy transfer and letting the voltage drift safely back down to the target baseline.

Because this measurement, comparison, and correction cycle repeats hundreds of thousands of times per second, the closed-loop system catches micro-volt deviations instantaneously. The result is a rock-solid, flat DC output voltage continuously delivered to the 2.4Ω load.

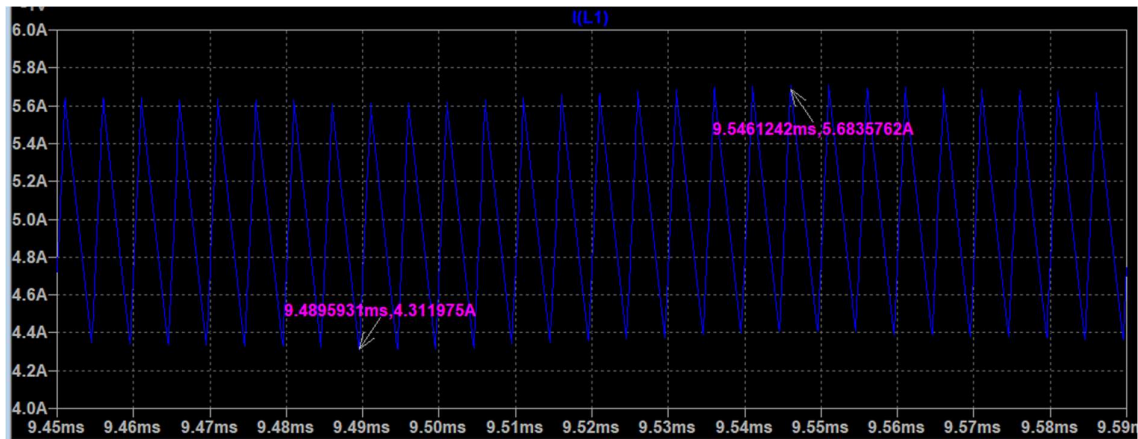
5. Simulation results and analysis:

5.1 Output voltage analysis V(n003): The simulated output voltage varies from peak 12.08V to 11.93V with an average voltage 12.008V. This results in a total peak to peak voltage ripple (ΔV_{out}) of $\sim 150\text{mv}$.

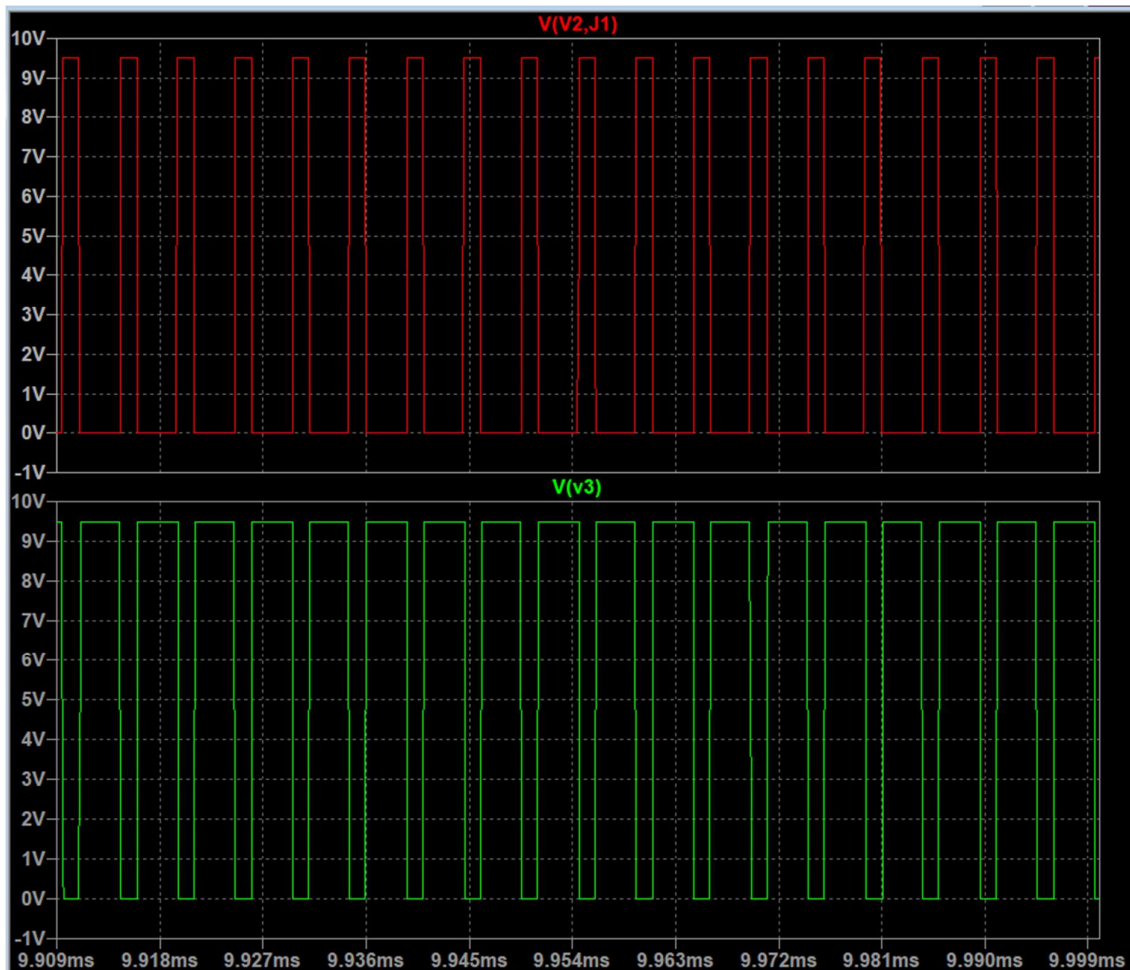


- **Waveform behaviour:** The output voltage is regulated around 12.008. A small high-frequency switching ripple is observed at the switching frequency of 200KHz. In addition, a low-frequency oscillatory component is present due to closed-loop control dynamics and compensator behaviour. This converter remains regulated and stable with an average output voltage close to the target value.

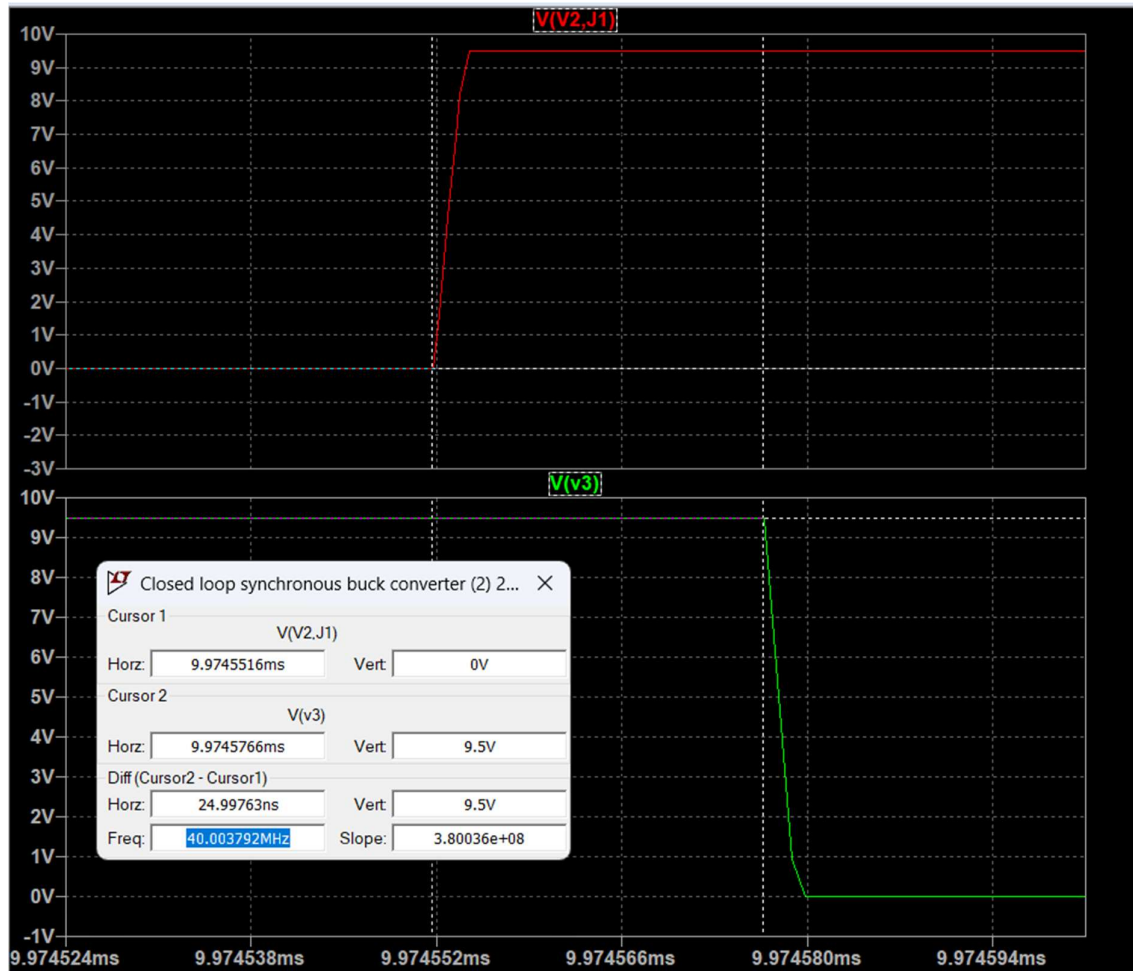
5.2 Inductor current analysis: The Converter operates in continuous conduction mode (CCM). The inductor current varies between approximately 4.312A to 5.683A with an average current of about 5.003A corresponding closely to the rated 5A load condition.



5.3 Gate drive timing verification: The gate-drive waveforms of the high-side $V(V2,J1)$ and low-side MOSFETs $V(v3)$ were analysed to verify proper synchronous switching operation of the converter. Complementary PWM signals were applied to both MOSFETs with a controlled dead-time to prevent simultaneous conduction and shoot-through current. The behavior were observed based on the structural logic of the circuit.

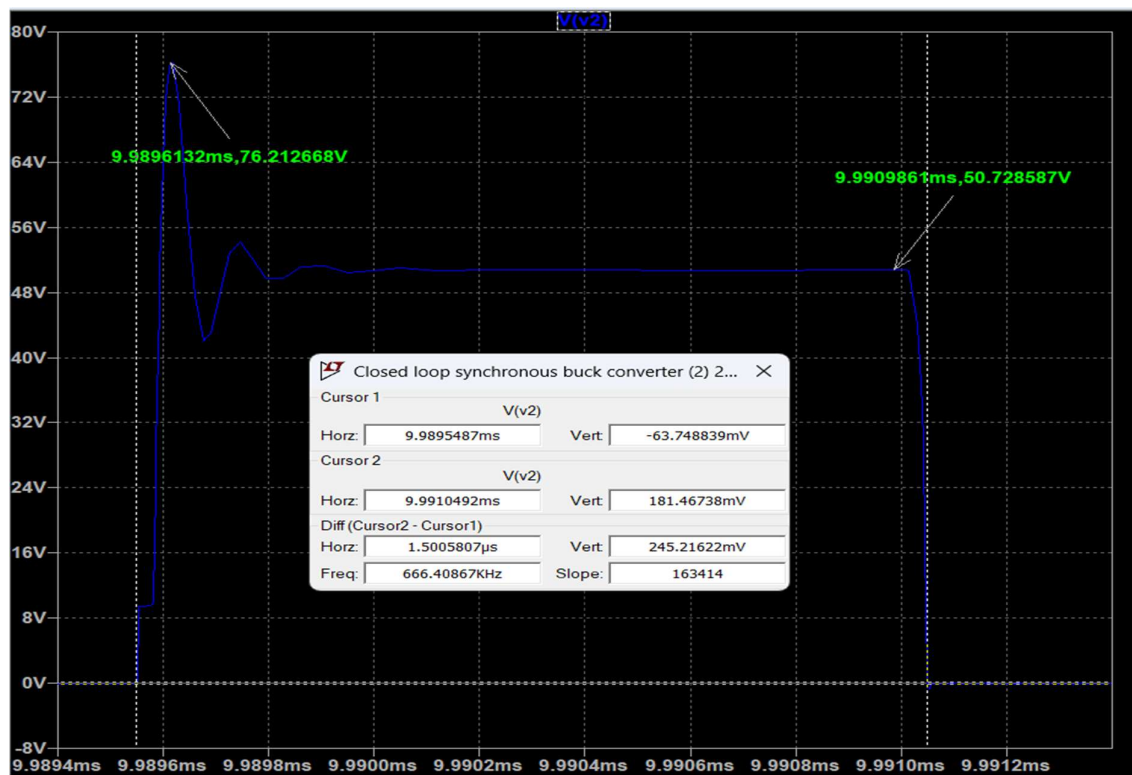
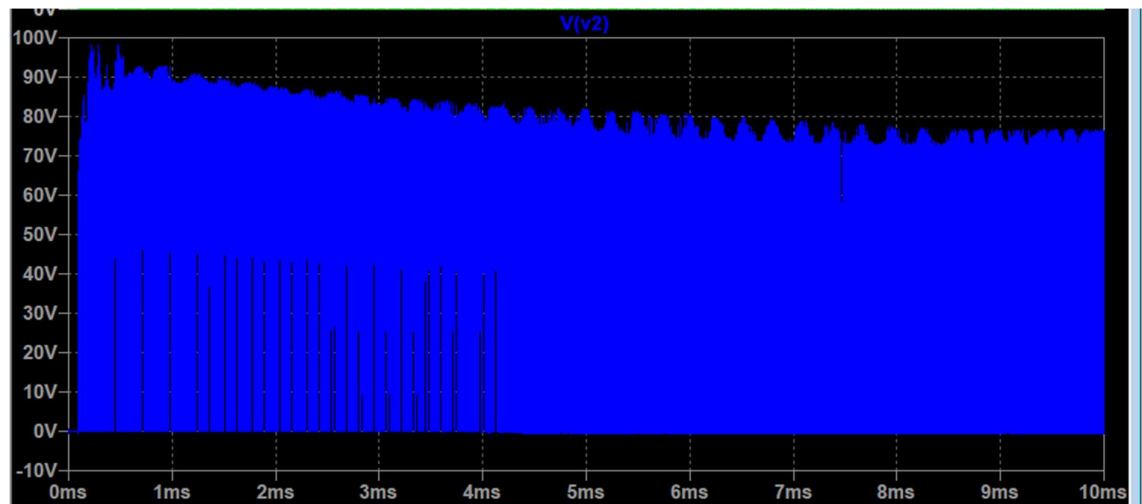


5.3.1 Command to Gate propagation delay: When tracking the Master command signal (PWM_OUT) against the high-side gate execution V(V2,J1), a discrete transport delay of exactly $\sim 25\text{ns}$ was captured via cursor analysis. This perfectly matches the ns delay parameter programmed into arbitrary behavioral source expressions to establish a safe propagation buffer.

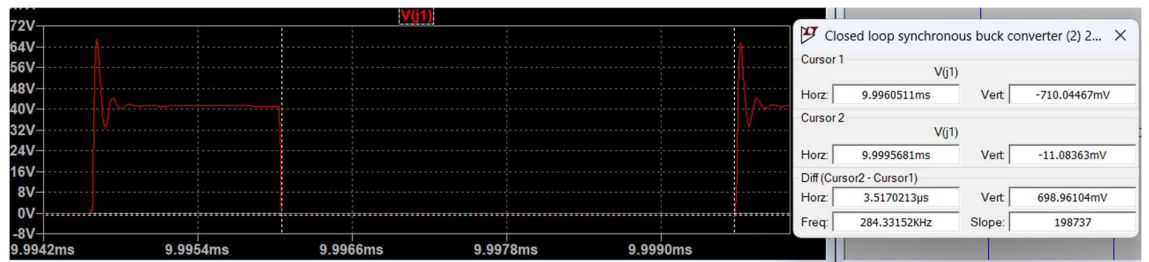
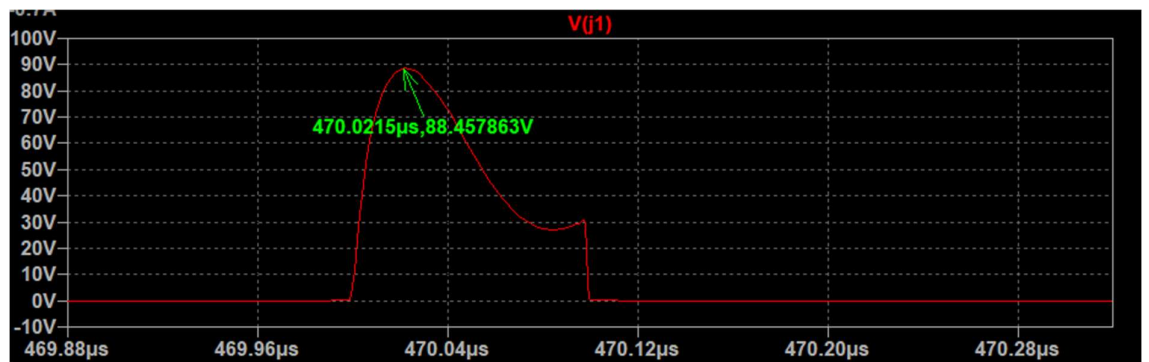
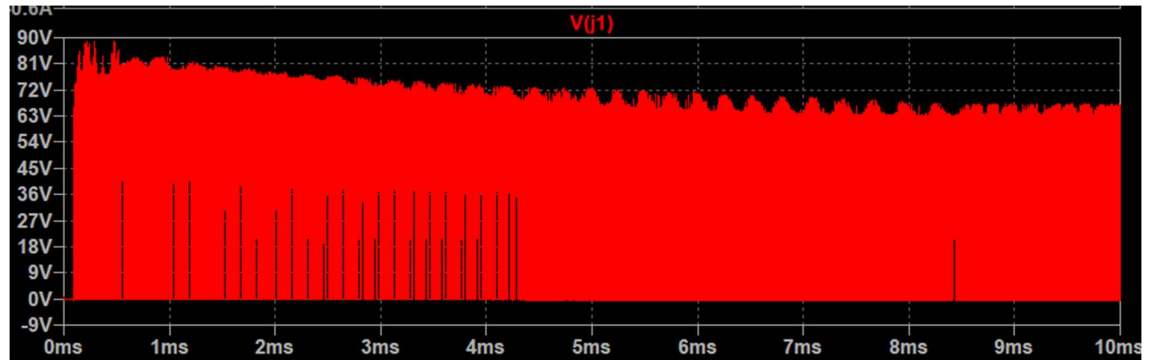


5.4 Switch Node Voltage Analysis: When the feedback loops are closed, the regulator actively modulates the pulse width command, altering the transient envelope and internal behavioral driver calculation matrix steps:

- **High side Gate driver node V(v2):** Closing the loop dampens the initial inrush path reducing the absolute peak startup spike to 97.940V. Due to higher dynamic current demands of the feedback loop the steady state localized edge overshoot shifts to a general maximum of 76.212V, while maintaining a stable flat-top conduction level of 50.728V. The on time of this node will be 1.500 μs and off time will be 3.454 μs .



- **Physical power switch node V(J1):** under active closed loop regulation, the absolute startup transient spike drops to 88.457V. The steady state edge overshoot is tightly limited to general maximum of 66.200V with a nominal regulated conduction flat top of 41.196V. The on time of this node will be 1.477 μ s and off time will be 3.517 μ s



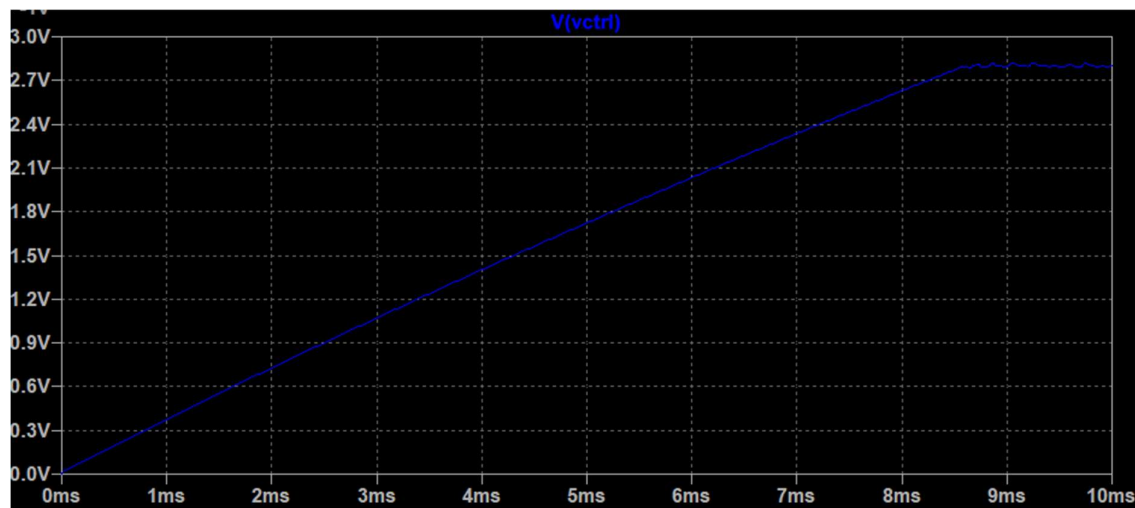
Observations:

- The physical switch node V(J1) and gate reference node V(V2) display stable, predictable square wave characteristics across both configurations.
- Closing the feedback loop successfully reduced absolute startup transient inrush spikes on both the gate line (down to 97.940V) and the switch node (down to 88.457V).
- The steady state flat top conduction levels perfectly validate the circuit metrics tracking at approximately 51V for the gate path and 41.4V for power switch path.
- The integrated RC snubber configurations efficiently damp high frequency parasitic tank energy, keeping all peak voltage overshoot transitions safely within the 120V semiconductor safety limit.

5.5 Control loop command V(vctrl): The error amplifier control voltage node V(vctrl) was analysed to evaluate loop stability, dynamic response range, and steady-state operating margins during the conversion cycle.

The control rail V(vctrl) represents the continuous analog command output generated by the loop error amplifier. This voltage level serves as the threshold boundary against which the high-frequency internal triangular carrier ramp is compared to generate the variable pulse-width switching commands. Tracking this node reveals the core regulatory action of the feedback loop:

- **Transient Startup Modulation Range:** During the initial power-on phase, the output capacitor filter is uncharged. The control loop actively driving the system from a cold start causes V(Vctrl) to adjust dynamically down to an empirical minimum of 2.789 V to safely manage transient initialization parameters.
- **Steady-State Regulation Settlement:** Once the output rail achieves complete target convergence, the feedback network stabilizes. Under full steady-state closed-loop regulation, the control voltage settles smoothly at 2.819 V.



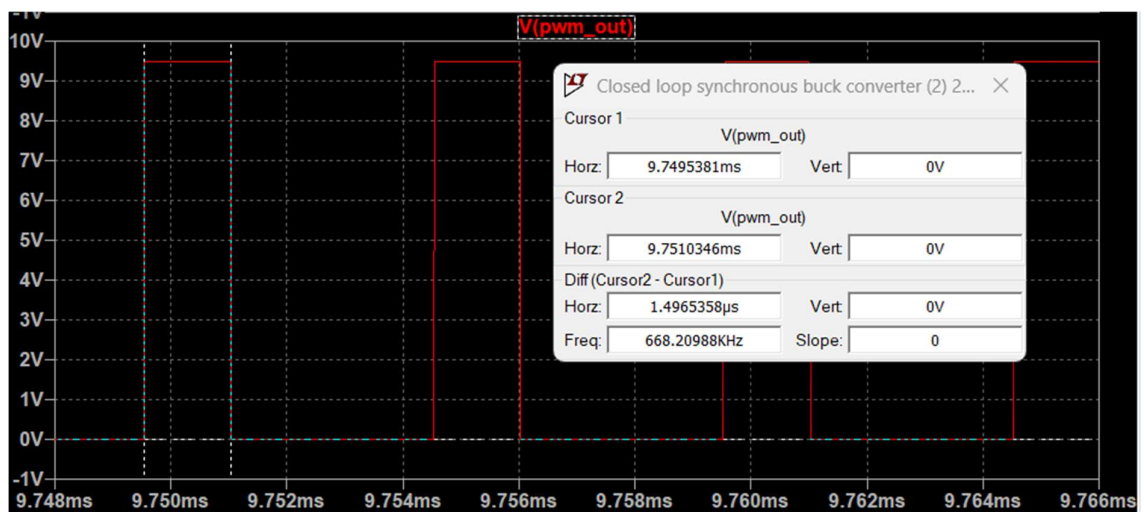
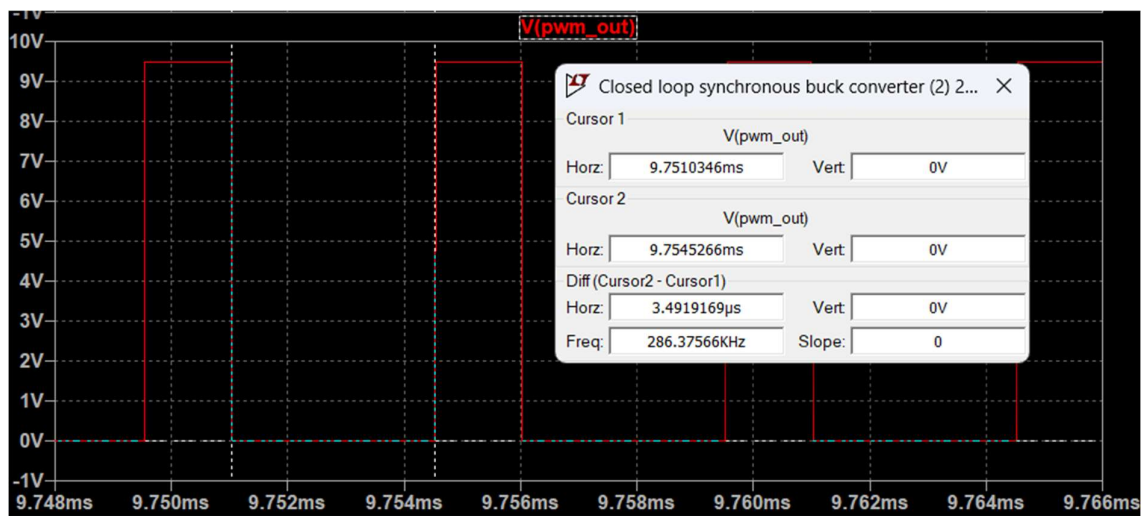
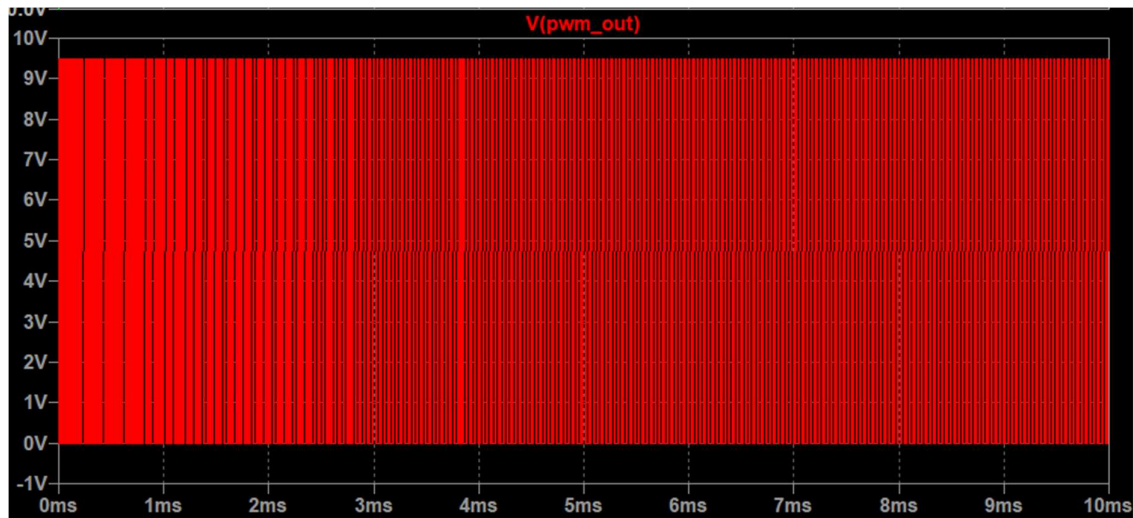


This narrow, stable voltage transition confirms that the modulator loop successfully controls duty cycle tracking without breaking into wide chaotic sweeps or low-frequency sub-harmonic oscillations, verifying baseline loop compensation.

Observations:

- The control voltage $V(Vctrl)$ operates with low voltage ripple, indicating a well-filtered error amplifier output stage.
- Dynamic tracking was verified across a clean operational sweep from 2.789 V during early transient startup up to 2.819 V at steady-state regulation.
- Stable settling behavior at 2.808 V confirms that the feedback network successfully calculates the correct steady-state duty cycle required by the switching matrix.
- No evidence of control loop hunting, oscillation, or pulse-width jitter was detected on the command rail.

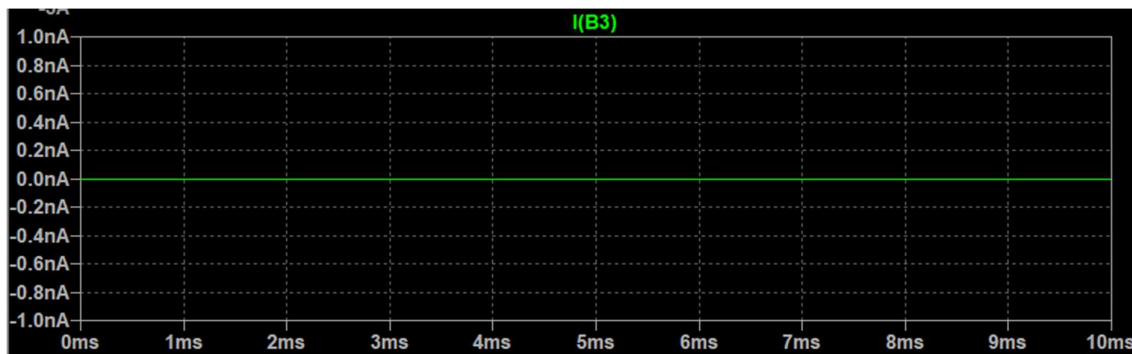
5.6 Pulse width Modulator Master logic output $V(PWM_OUT)$: The modulator output operates as a clean noise free master digital pulse train toggling sharply between a true logic low of 0V and a stable logic high level of 9.5V. Throughout the observed timeline, $V(pwm_out)$ functions as a highly uniform periodic square wave because this wave is calculated through behavioral comparator modelling, it switches instantaneously with zero localized transition edge ringing or signal tailing. Cursor measurements of the raw comparator output track an active conduction pulse width(T_{on}) of exactly 1.496 μ sec and a logic low off-time duration(T_{off}) of 3.491 μ sec. Combining these periods yields a total fundamental switching.($T_{period} = T_{on} + T_{off}$) of 4.987 μ sec, establishing an internal carrier reference of 200.8KHz at a master duty cycle of: 0.3012 comparing this duty cycle with final high side gate driver pulse width (1.496 μ sec) verifies the successful insertion of dead time traking.



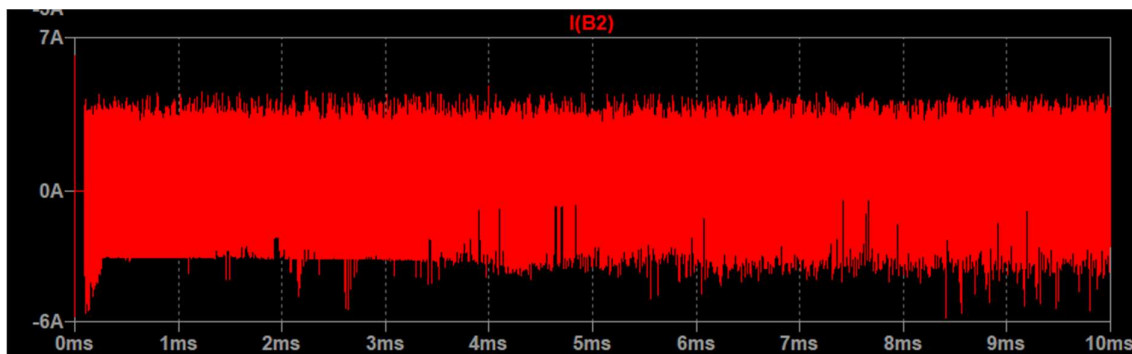
5.7 Behavioral Source Current Analysis (IB1, IB2, and IB3): The dynamic current profiles flowing from the behavioral logic blocks IB1, IB2, and IB3 were evaluated to determine peak transient loading capacity and define the physical hardware drive requirements.

While node voltages confirm logical timing states, analysing the current waveforms exposes the dynamic power delivery requirements of the control architecture. In real hardware, gate driver networks must supply high-amplitude current pulses to rapidly charge and discharge parasitic internal capacitances during fast switching transitions (di/dt). The behavior of these three distinct current nodes details the drive stage loading characteristics:

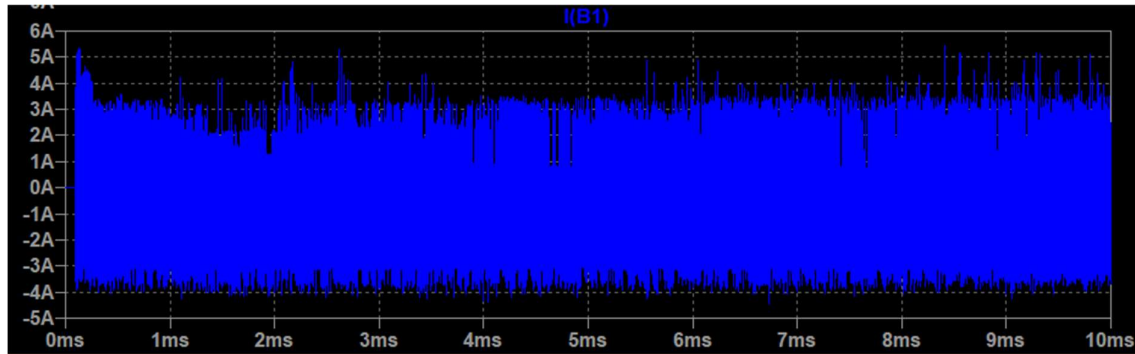
5.7.1 Master Logic Generation Current I(B3): The current profile for the master PWM generator node tracks flatly at 0.0 nA. Because block B3 acts strictly as an ideal mathematical signal processing logic cell rather than a power driver, its current consumption is negligible. This confirms clear structural separation between control signal evaluation and power delivery.



5.7.2 High-Side Drive Current I(B2): This trace captures the intense dynamic current bursts demanded to toggle the high-side power switch. During switching transitions, I(B2) exhibits high-frequency transient pulses swinging continuously between an upper sourcing peak of +3.5 A and a sharp sinking trough of -5.4 A. These boundaries define the exact current delivery capacity needed to overcome high-side gate charge thresholds rapidly.



5.7.3 Low-Side Drive Current I(B1): Operating as the complementary gate-drive controller, I(B1) displays a massive transient envelope tracking between a peak sourcing current of +6 A and a sinking limit of -3 A. This high sourcing current ensures rapid enhancement of the low-side synchronous switch channel, clamping conduction losses down to ideal levels.



Engineering Significance for Hardware Selection

In a physical converter design, these simulated current spikes dictate the selection of the driver IC's output stage. The peak positive and negative current boundaries captured across I(B2) (+4A / -5A) and I(B1) (+6A / -3A) ensure that the chosen hardware gate driver can source and sink sufficient transient current to minimize switching losses without experiencing thermal overstress or internal bond-wire degradation.

Observations

- The current waveforms I(B2) and I(B1) consist of highly symmetric, periodic transient pulses corresponding directly to the switching transitions of the power stage.
- Peak sourcing and sinking current spikes occur instantaneously at the transition edges, validating that the behavioral blocks simulate realistic gate charge physics.
- The absolute zero baseline of the primary modulation current I(B3) confirms a clean, low-noise signal processing interface within the core behavioral error network.
- The absence of prolonged current trailing or erratic oscillations proves that the gate drive network achieves rapid, clean transition clamping.

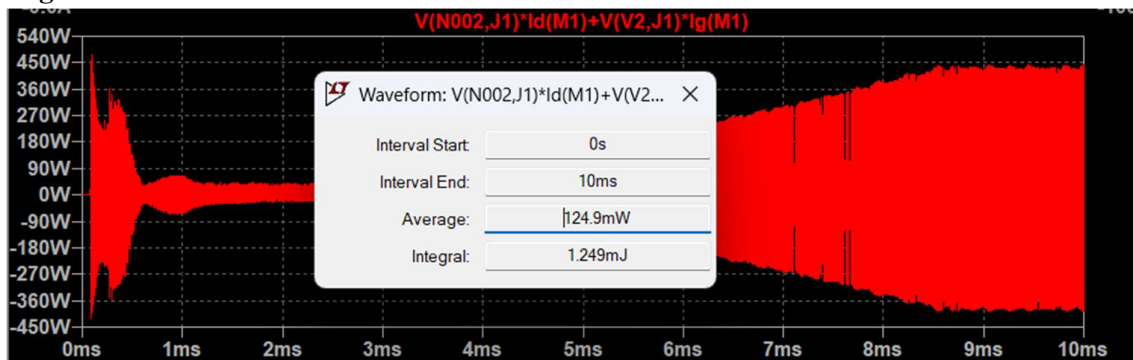
5.8 Efficiency calculation: The converter efficiency is calculated by comparing average output power to average input power. Output power is determined from the product of average output voltage and load current while input power is determined from the product of average input voltage and source current. The calculated efficiency reflects the cumulative impact of the conduction losses, parasitic resistance, diode forward drop, and switching losses in the practical converter model.

$$\eta(\%) = \frac{P_{out}}{P_{in}} = \frac{V_{out(avg)} * I_{out(avg)}}{V_{in(avg)} * I_{in(avg)}} * 100 = \frac{12.008 * 5.003}{41.851 * 1.486} * 100 = \frac{60.076}{62.190} * 100 = 96.60\%$$

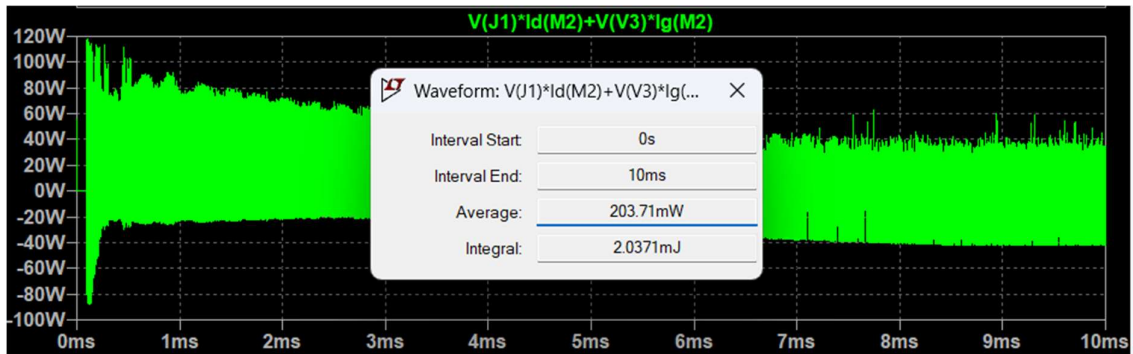
```
vin_avg: AVG(V(n001) )=41.851341945 FROM 0.0095 TO 0.01
iin_avg: AVG(I(V1) )=-1.48658059469 FROM 0.0095 TO 0.01
vout_avg: AVG(V(n003) )=12.0080034001 FROM 0.0095 TO 0.01
iout_avg: AVG(I(R1) )=5.00333475274 FROM 0.0095 TO 0.01
```

6. Loss component analysis:

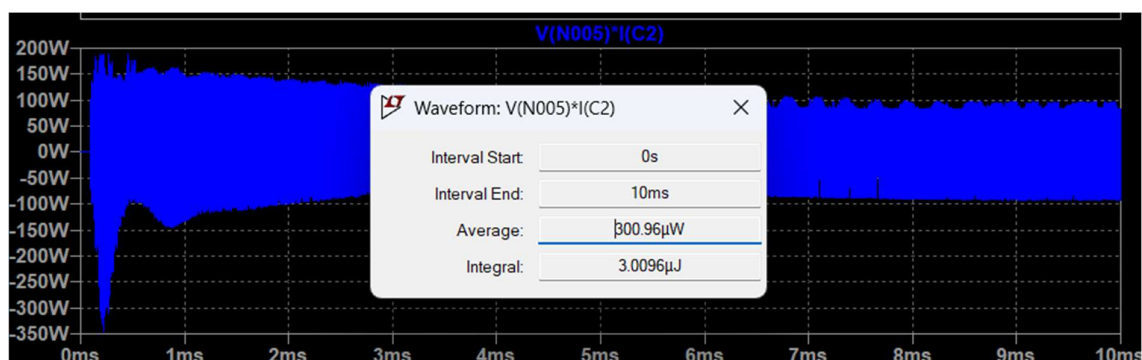
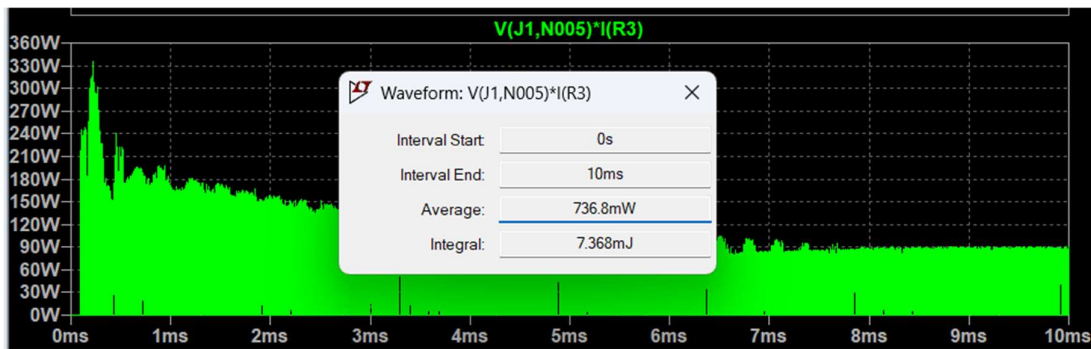
6.1 High side MOSFET Conduction loss: 124.9mw



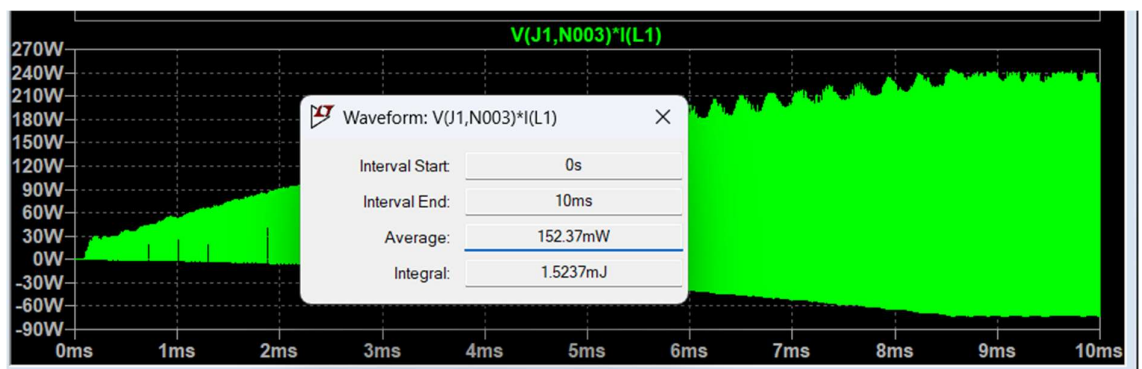
6.2 Low side MOSFET Conduction loss: 203.71mw



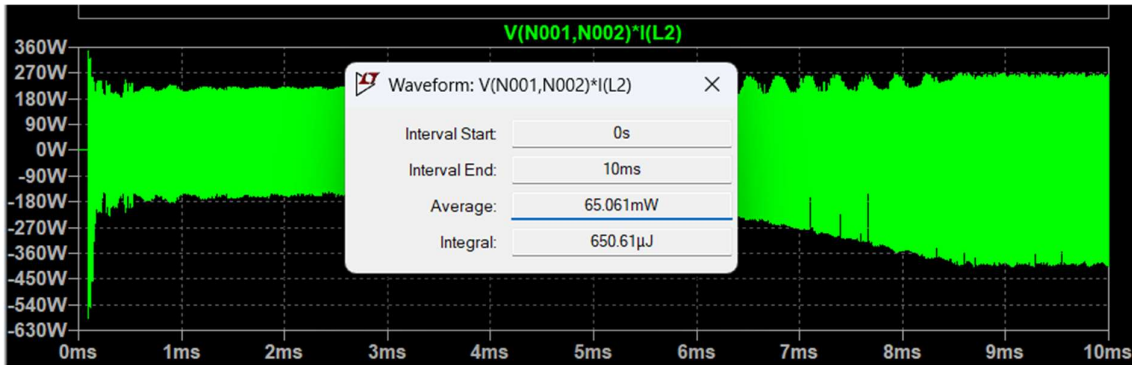
6.3 Snubber losses: R_{loss} : 736.8mw, C_{loss} : 300.96 μ w



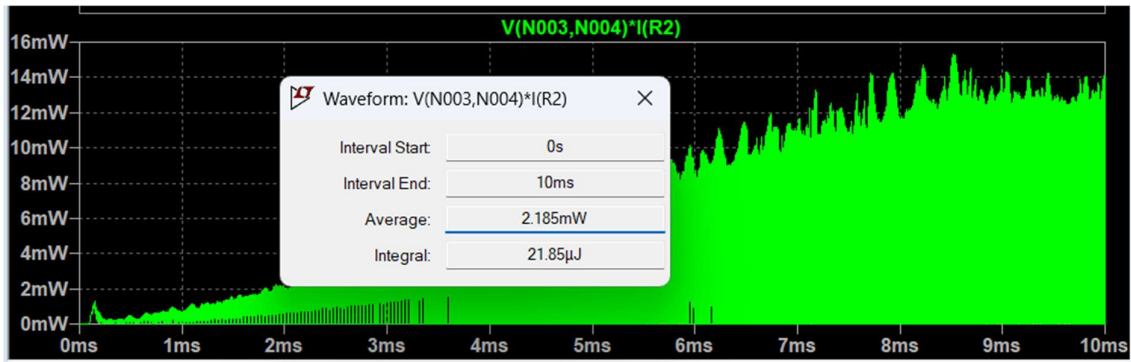
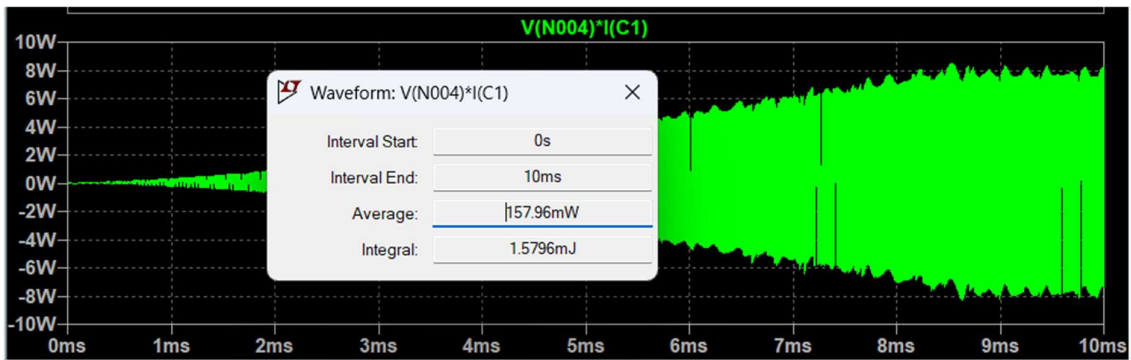
6.4 Inductor loss(Core): 152.37mw



6.5 Input inductance loss: 65.061mw



6.6 Output capacitance loss (ESR): R_{loss} : 2.185mw, C_{loss} : 157.96mw



6.7 Comprehensive power loss breakdown table:

Loss component	Mathematical derivation/Formula	Monitored/Calculated parameter	Loss(W)
High Side MOSFET Conduction loss	$I_{Avg}^2 * R_{DS(on)} * D$	From simulation plot window	0.1249
Low-side MOSFET conduction loss	$I_{Avg}^2 * R_{DS(on)} * (1-D)$	From simulation plot window	0.20371
Switching loss	$0.5 * V * I * (tr + tf) * f$	$0.5 * 42 * 5 * (2.35n + 3.95n) * 200K$	0.1745
Inductor copper loss (DCR)	$I_{Avg}^2 * DCR$	$5^2 * 0.01$	0.25
Source resistance loss	$I_{In(avg)}^2 * R_S$	$1.486^2 * 0.1$	0.22
Inductor loss (Core)	Integrated Core Power Trace: $\frac{1}{T} \int (V_L \cdot I_L) dt$	From simulation plot window	0.15237
RC Snubber loss	Integrated Resistor Power Trace: $\frac{1}{T} \int (V_R \cdot I_R) dt$	From simulation plot window ($R_{loss} + C_{loss}$)	0.73710
Output capacitance loss (ESR)	Integrated Capacitor ESR Power Trace	From simulation plot window ($R_{loss} + C_{loss}$)	0.16014
Input inductance loss	Integrated Input Filter Power Trace	From simulation plot window	0.065601
Total calculated losses	-	-	2.0883
Miscellaneous Parasitic Losses	Residual losses attributed to behavioral sources, compensator system parasitic, gate drive effects and unmodeled parasitics etc.	-	~ 0.0257
Total System Losses	$\Sigma P_{losses} = P_{in(avg)} - P_{out(avg)}$	$P_{in} = 62.190 \text{ W}, P_{out} = 60.076 \text{ W}$	2.114

6.8 Loss Model Validation and Analysis: To verify the mathematical and physical accuracy of the synchronous buck converter design, a dual-verification power balance methodology was implemented. The overall global system power dissipation was extracted directly from the LTspice simulation by tracking the average input power delivered from the source and subtracting the average power consumed by the load:

$$P_{\text{loss(global)}} = P_{\text{in(avg)}} - P_{\text{out(avg)}} = 62.190 \text{ W} - 60.076 \text{ W} = 2.114 \text{ W}$$

Concurrently, a discrete bottom-up loss model was constructed by summing the individually isolated component losses derived from transient waveform integrations and analytical circuit equations (as detailed in Table 6.7). The summation of these discrete localized losses yields:

$$P_{\text{loss(calculated)}} = \text{Sum of Losses} = 2.0883 \text{ W}$$

Comparing the two methods reveals a minute discrepancy of exactly 0.0257 W (25.7mw), representing an exceptionally tight alignment of 98.78% between the analytical model and the global simulator parameters. This minor residual variance is classified under miscellaneous parasitic losses, which physically account for:

- **Gate-Charge Distribution Losses (Qg):** Shifting energy through the finite internal gate mesh resistances during turn-on and turn-off transients.
- **Body-Diode Conduction:** Minor conduction intervals of the low-side synchronous MOSFET's internal body diode during the dead-time deadband transitions before the channel fully enhances.

➤ Key Loss Insights:

- **Dominant Losses:** The RC snubber network stands out as the largest single loss contributor (0.73710 W). This is an expected hardware trade-off, as the snubber intentionally dampens high-frequency parasitic layout ringing to protect the high-side switch from overvoltage spikes.
- **Conduction Symmetry:** The low-side MOSFET conduction loss (0.20371 W) is significantly higher than that of the high-side switch (0.1249 W). This cleanly validates the converter's step-down ratio; because the duty cycle is low, the synchronous low-side FET conducts current for the majority of the switching period, naturally scaling up its power dissipation based on the low-side root-mean-square current squared multiplied by its on-resistance ($I_{\text{rms(LS)}}^2 * R_{\text{ds(on)}}$).

7 Key Learnings

- The development of the Closed-Loop Synchronous Buck Converter (CLSBC) provided practical understanding of converter design, component selection, closed-loop control implementation, and performance verification through simulation.
- The project demonstrated the relationship between power stage parameters and output voltage regulation, highlighting the importance of proper selection of switching frequency, inductor, capacitor, and feedback network components to achieve the desired performance.
- Through closed-loop operation, it was observed that the converter maintained regulated output voltage under varying operating conditions, validating the effectiveness of the control strategy and overall design approach.
- The study also provided experience in analyzing key power electronics performance parameters, including output voltage regulation, inductor current behavior, power loss estimation, efficiency evaluation, and converter stability assessment.
- Overall, the project strengthened practical knowledge of DC-DC converter design, simulation-based validation, and engineering analysis.

8 Future Scope

- The current work demonstrates the design and simulation of a Closed-Loop Synchronous Buck Converter (CLSBC) with regulated output voltage operation. As a continuation of this project, the converter can be validated across a multiple load conditions further to assess regulation performance and operating limits under practical scenarios in version 2.
- Further work will include frequency-domain analysis through Bode plot evaluation to verify the dynamic behavior of the control system and ensure satisfactory performance over the intended operating range.
- In addition, load transient analysis will be performed to evaluate the converter's response to sudden variations in load current and to assess key performance parameters such as output voltage deviation and recovery characteristics also will test this converter at various input voltage.
- The project can also be extended to hardware implementation, including PCB design, component selection, efficiency evaluation, thermal analysis, and

experimental validation. These enhancements will provide a comprehensive understanding of converter design and verification practices commonly followed in industrial power electronics applications.

9 Conclusion

A Closed-Loop Synchronous Buck Converter (CLSBC) was successfully designed and simulated to achieve a regulated DC output voltage. The converter demonstrated stable operation and satisfactory performance under nominal steady-state operating conditions. Key parameters such as output voltage, inductor current, efficiency, and power losses were evaluated to validate the baseline design. While these results confirm the steady-state effectiveness of the proposed architecture, real-world deployment demands rigorous dynamic testing. Therefore, comprehensive evaluations involving load transient steps, multiple loading scenarios, and frequency-domain stability (Bode plot) analysis will be executed and detailed in Version 2 of this design report to bridge the gap between simulation and hardware reality.